Using Bilingual Responses to Enhance Cultural Knowledge in AI

Joshitha Reddy Bangaru, B24DS009 and Aarush Ranjan, B24DS002

***Abstract*—Large Language Models (LLMs) have demonstrated significant capabilities in natural language understanding and generation. However, their ability to accurately represent everyday cultural knowledge remains inconsistent across different countries, regions, and languages. This limitation is particularly evident for underrepresented cultures and low-resource languages, where models may generate generic, incomplete, or culturally inaccurate responses.**

Recent research, particularly the BLEnD benchmark, highlights significant disparities in the cultural knowledge of LLMs and demonstrates that answer quality can vary depending on the language used for prompting. Motivated by these findings, this project investigates a dual-language response generation framework that utilizes both local-language and English responses to improve cultural question answering.

The proposed approach combines multilingual response generation, semantic comparison, Retrieval-Augmented Generation (RAG), and Large Language Model-as-a-Judge (LLM-as-a-Judge) to identify, verify, and select more reliable answers. A structured Cultural Knowledge Base is incorporated to provide external evidence during answer evaluation.

Furthermore, the implementation addresses practical limitations in cultural retrieval systems, including cross-cultural semantic interference, inference latency, lack of source attribution, and geographic representation bias. The overall objective is to improve cultural accuracy, reduce language-dependent performance disparities, and produce more evidence-grounded responses across diverse cultural contexts.

***Index Terms*—Large Language Models, Cultural Knowledge, Multilingual NLP, BLEnD, Retrieval-Augmented Generation, LLM-as-a-Judge, Cross-Lingual Question Answering, Semantic Similarity, Knowledge Retrieval.**

I. Project Details

A. Project Criterion

- **Criterion: Existing Code and Research Extension**
  The project is based on the publicly available BLEnD benchmark and its associated research resources. The work focuses on extending existing cultural question-answering approaches through a bilingual ensemble framework and knowledge-base-assisted answer evaluation.
- **Research Limitation:**
  BLEnD identifies that LLMs may perform differently when answering cultural questions in local languages and English. However, its primary objective is to evaluate and measure these differences rather than provide an inference-time mechanism for using language-dependent performance differences to improve the final answer.
  Additionally, conventional single-response approaches may return culturally inaccurate or generalized information without verifying the generated answer against an external cultural knowledge source.
- **Proposed Solution:**
  We propose a dual-language ensemble framework that generates responses in both the local language and English.
  Instead of directly accepting either response, the framework treats both outputs as complementary candidates. A semantic comparison mechanism identifies their level of agreement or disagreement.
  When necessary, relevant cultural information is retrieved from an external Cultural Knowledge Base using Retrieval-Augmented Generation (RAG).
  An LLM-as-a-Judge module subsequently evaluates the generated responses using the retrieved evidence and selects or synthesizes a more reliable final answer.
  The implementation further incorporates metadata-based retrieval filtering, in-memory model caching, structured fact provenance, and geographic dataset expansion to address practical limitations in cultural question-answering systems.

B. Team Composition and Individual Contributions

- **Member 1: Joshitha Reddy Bangaru**
  Roll Number: B24DS009
  Responsible for developing the dual-language ensemble framework, implementing local-language and English response generation, designing the answer comparison mechanism, and integrating the LLM-as-a-Judge adjudication process with knowledge-base-assisted decision making.
- **Member 2: Aarush Ranjan**
  Roll Number: B24DS002
  Responsible for constructing and curating the cultural knowledge base, designing the retrieval strategy for supplying evidence to the judge model, and conducting evaluation, error analysis, and visualization of accuracy improvements across countries.

C. Research Motivation

The motivation behind this project is that multilingual capability does not necessarily guarantee equivalent cultural understanding across languages.

A model may possess relevant cultural information but fail to retrieve or express it effectively when prompted in a particular language.

Generating responses in multiple languages provides an opportunity to examine different knowledge representations available within the same model.

The proposed framework further introduces external cultural evidence to support answer verification, rather than relying entirely on the model's internal knowledge.

D. Research Objectives

The primary objectives of this project are:

- To investigate the impact of language selection on the accuracy of culturally dependent responses.
- To develop a bilingual response generation mechanism using local-language and English prompts.
- To identify semantic agreement and disagreement between responses generated in different languages.
- To construct a structured Cultural Knowledge Base containing country-specific information.
- To incorporate Retrieval-Augmented Generation for retrieving supporting cultural evidence.
- To develop an LLM-as-a-Judge mechanism for evidence-based response selection and synthesis.
- To improve retrieval reliability through country-specific metadata filtering.
- To evaluate the proposed approach against individual language-based baselines.

E. Key Contributions

The project focuses on the following contributions:

1) **Dual-Language Response Generation:** Generating two independent responses using local-language and English versions of the same cultural question.
2) **Cross-Lingual Response Comparison:** Evaluating semantic similarity and identifying disagreements between bilingual responses.
3) **Cultural Knowledge Integration:** Incorporating an external knowledge repository to provide relevant cultural facts during answer verification.
4) **Evidence-Grounded Adjudication:** Utilizing an LLM-as-a-Judge to evaluate candidate responses using both generated answers and retrieved supporting information.
5) **Metadata-Scoped Retrieval:** Applying country-specific metadata filtering to reduce retrieval of culturally unrelated information.
6) **Efficient Inference Infrastructure:** Using an in-memory FastAPI service to cache the embedding model and retrieval index.
7) **Structured Fact Provenance:** Maintaining source information and metadata for retrieved cultural knowledge.
8) **Geographic Expansion:** Expanding the cultural knowledge collection to include additional geographically diverse countries and underrepresented regions.

II. Problem Statement

A. Context and Motivation

Large Language Models are increasingly used for information retrieval, conversational systems, question answering, and multilingual communication. Despite their impressive capabilities, these models do not consistently demonstrate an accurate understanding of everyday cultural practices and traditions across different communities.

Cultural knowledge includes various aspects of everyday life, such as food habits, festivals, educational practices, family traditions, sports, celebrations, and social customs.

Unlike general factual knowledge, many cultural practices are region-specific and may not be sufficiently represented in publicly available digital resources.

Consequently, LLMs may generate responses that are factually incorrect, overly generalized, or influenced by cultural information from other regions.

These limitations can affect the reliability and inclusiveness of AI systems, particularly when interacting with users from underrepresented linguistic and cultural communities.

The BLEnD benchmark introduced by Myung et al. [1] provides an important foundation for studying this problem.

It contains approximately 52.6K question-answer pairs covering 16 countries and regions across 13 languages. The benchmark includes both Short-Answer Questions (SAQ) and Multiple-Choice Questions (MCQ).

The study demonstrates that LLM performance varies significantly across cultural contexts. It also identifies a language-dependent performance pattern: models tend to perform better in local languages for cultures associated with mid-to-high-resource languages, whereas English responses can perform better for cultures associated with low-resource languages.

These observations motivate the need for an approach that does not rely exclusively on a single language when answering culturally dependent questions.

B. Why This Problem Is Important in Machine Learning

Cultural knowledge representation is an important challenge in multilingual artificial intelligence.

Improving the reliability of cultural question answering can contribute to:

- **Fairness:** Reducing differences in AI response quality across languages and cultures.
- **Reliability:** Reducing culturally inaccurate or unsupported responses.
- **Inclusiveness:** Improving representation of underrepresented communities.
- **Generalization:** Improving the ability of models to respond across different cultural contexts.
- **Trustworthiness:** Providing external evidence to support generated cultural information.

C. Specific Scope of the Work

The project focuses on improving cultural question answering through a dual-language ensemble and knowledge-base-assisted adjudication framework.

For every cultural question:

1) The original question is processed in its local-language version.
2) The corresponding English version of the question is also provided to the LLM.

3) The system generates two candidate answers.
4) The answers are compared using semantic similarity and disagreement detection.
5) Relevant cultural evidence is retrieved from the Cultural Knowledge Base.
6) An LLM-as-a-Judge evaluates the candidate answers using the retrieved evidence.
7) The system selects or synthesizes the final response.

The scope also includes improving retrieval efficiency, maintaining source attribution, and examining geographic representation in the cultural knowledge collection.

The primary focus is inference-time improvement rather than training a new language model.

III. Methodology

A. Proposed ML Models and Architecture

The proposed system consists of multiple interconnected components.

1) Multilingual Large Language Model: A multilingual LLM is used to generate responses to the same cultural question in two languages:

- Local-language response generation.
- English response generation.

Both responses are treated as independent candidate answers for subsequent evaluation.

2) Dual-Language Ensemble: The dual-language ensemble combines the information generated through both language representations.

Rather than selecting an answer solely based on the language in which it was generated, the framework compares both candidates before making a final decision.

3) Semantic Similarity and Disagreement Detection: Sentence embeddings are used to represent the generated answers in a common vector space.

Cosine similarity is calculated as:

$$\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\|\|B\|} \tag{1}$$

where A and B represent the embedding vectors of the local-language and English responses.

The similarity score provides an initial indication of semantic agreement.

An additional LLM-based comparison mechanism can classify the responses as:

- Agree
- Partially Agree
- Disagree

Semantic similarity is treated as a comparison signal rather than proof of factual correctness.

4) Cultural Knowledge Base: A structured Cultural Knowledge Base is maintained to provide external information about different countries and regions.

The knowledge repository contains information related to:

- Traditional food and food habits.
- Festivals and celebrations.
- Educational and school customs.
- Family traditions.
- Social practices.
- Local cultural activities.
- Everyday cultural information.

Each knowledge entry is associated with relevant metadata, including country, category, topic, and source information.

5) Retrieval-Augmented Generation: Retrieval-Augmented Generation is used to retrieve relevant cultural information before the final answer is adjudicated.

The retrieval process consists of:

1) Converting cultural questions and knowledge entries into vector embeddings.
2) Searching the vector index for semantically relevant candidates.
3) Applying country-specific metadata filtering to restrict the retrieved information to the target cultural context.
4) Selecting the most relevant evidence for the final adjudication process.

The framework uses FAISS for vector similarity search.

6) LLM-as-a-Judge: The LLM-as-a-Judge module evaluates the two candidate answers using:

- Original cultural question.
- Local-language generated answer.
- English generated answer.
- Retrieved cultural evidence.

The judge determines which response is better supported by the available evidence.

If neither response is sufficiently accurate, the judge may synthesize a more appropriate answer using the retrieved information.

The objective is to make the final decision evidence-grounded rather than relying only on the fluency of generated responses.

B. Overall System Architecture

The proposed architecture follows the pipeline illustrated in Fig. 1.

The overall workflow is:

1) BLEnD Question Input.
2) English and Local-Language Prompt Generation.
3) Multilingual LLM Response Generation.
4) Local and English Answer Comparison.
5) Semantic Agreement and Disagreement Detection.
6) Cultural Knowledge Base Retrieval using RAG.
7) Country-Specific Metadata Filtering.
8) LLM-as-a-Judge with Retrieved Evidence.
9) Final Answer Selection or Synthesis.
10) Accuracy Evaluation and Cultural Gap Analysis.

C. Key Techniques, Algorithms and Frameworks

The following techniques are incorporated into the proposed system.

TABLE I
MAJOR TECHNIQUES USED IN THE PROPOSED FRAMEWORK

Technique	Purpose
Multilingual Prompting	Generate responses in local language and English.
Sentence Embeddings	Convert text into numerical vector representations.
Cosine Similarity	Measure semantic similarity between responses.
FAISS	Perform efficient vector similarity search.
RAG	Retrieve external cultural evidence.
Metadata Filtering	Restrict retrieval to the relevant country and context.
LLM-as-a-Judge	Evaluate, select, or synthesize candidate responses.
FastAPI	Expose the pipeline through an API service.
SQLite	Store questions, answers, decisions, and evaluation records.

D. Difference from Existing BLEnD Approach

BLEnD is primarily a benchmark and evaluation framework that measures the everyday cultural knowledge of LLMs across different countries and languages.

It identifies that the choice of language can influence the accuracy of generated cultural responses.

However, the proposed project uses this observation as a basis for an answer-improvement mechanism.

The distinction is summarized below:

- **BLEnD:**
  Local-language response versus English response $\rightarrow$ Measure performance differences.
- **Proposed Framework:**
  Local-language response and English response $\rightarrow$ Compare both responses $\rightarrow$ Retrieve cultural evidence $\rightarrow$ LLM-as-a-Judge $\rightarrow$ Final answer.

The primary distinction is that the proposed framework attempts to actively utilize cross-lingual differences to improve answer selection rather than only measuring them.

E. Additional Research Gaps and Implemented Improvements

Beyond the bilingual response framework, the implementation addresses four practical limitations identified during the study of cultural retrieval systems.

1) Cross-Cultural Semantic Bleed: **Research Limitation:**

Standard dense retrieval systems may retrieve semantically similar information from different countries, even when the cultural context of the question is specific to one country.

This can introduce irrelevant cultural facts into the answer-generation process.

Implemented Improvement: Over-Sampled Metadata Scoping

The FAISS retrieval process is configured to initially retrieve a larger candidate set of $k = 1000$ entries.

The retrieved candidates are subsequently filtered using country-specific metadata to restrict the final evidence to the target cultural context.

This approach is designed to reduce cross-country retrieval interference.

2) Prototype Latency and Model Overhead: **Research Limitation:**

Research prototypes that reload embedding models and retrieval indexes for individual requests introduce unnecessary computational overhead and increased response latency.

Implemented Improvement: In-Memory FastAPI Microservice

The system uses a FastAPI-based service that loads the multilingual-e5-large embedding model and FAISS index into memory during server startup.

The loaded components are reused across subsequent requests rather than repeatedly initialized.

The Phase 2 prototype presentation reports a reduction in query response time from approximately 15–20 seconds to under 100 milliseconds.

This latency figure should be interpreted with respect to the measured retrieval component and its test configuration, rather than the complete LLM generation pipeline.

3) Lack of Ground-Truth Attribution: **Research Limitation:**

Many retrieval-augmented question-answering systems generate responses without maintaining clear source-level attribution for individual cultural facts.

This makes it difficult to verify where retrieved information originated.

Implemented Improvement: Structured Fact Provenance

Each knowledge-base entry is associated with a structured metadata representation containing fields such as:

- Country.
- Cultural category.
- Topic.
- Fact description.
- Source URL.
- Relevant keywords.

This structure allows retrieved information to be traced back to its corresponding source record and source URL.

4) Geographic Data Bias: **Research Limitation:**

Cultural question-answering resources may provide uneven geographic representation, with some regions receiving substantially more attention than others.

This can limit the generalization of cultural retrieval systems.

Implemented Improvement: Global South Dataset Expansion

The cultural knowledge collection was expanded through additional Wikipedia-based information gathering across seven diverse countries.

The expansion includes underrepresented regions such as Ethiopia and Egypt.

This additional collection is intended to examine retrieval performance across a broader range of cultural contexts.

F. System Implementation Pipeline

The implementation follows the following logical sequence:

Algorithm 1 Dual-Language Cultural QA Framework

Require: Cultural question Q, target country C
Ensure: Final answer A_f
Generate local-language question Q_l
Generate English question Q_e
Obtain local answer A_l
Obtain English answer A_e
Compute semantic similarity $S(A_l, A_e)$
Identify agreement or disagreement
if Further verification is required **then**
    Retrieve candidate cultural facts
    Apply country-specific metadata filtering
    Select relevant supporting evidence
end if
Provide both answers and evidence to LLM Judge
Evaluate factual support and consistency
Select or synthesize final answer A_f
return A_f

IV. Resource and Dataset Availability

A. Dataset Description

The primary dataset used in this project is BLEnD (A Benchmark for LLMs on Everyday Knowledge in Diverse Cultures and Languages) [1].

- **Dataset Name:** BLEnD.
- **GitHub Repository:** https://github.com/nlee0212/BLEnD
- **Hugging Face Dataset:** https://huggingface.co/datasets/nayeon212/BLEnD
- **Dataset Size:** Approximately 52.6K question-answer pairs.
- **Geographic Coverage:** 16 countries and regions.
- **Language Coverage:** 13 languages.
- **Question Types:** Short-Answer Questions (SAQ) and Multiple-Choice Questions (MCQ).

B. Dataset Structure

The BLEnD dataset contains culturally grounded questions and corresponding answer information.

The relevant dataset components include:

- Cultural question.
- Local-language question.
- Corresponding English question.
- Country and regional information.
- Reference answers.
- Human annotation and evaluation information.

The dataset provides parallel question representations, enabling comparison of responses generated using different language prompts.

C. Data Preprocessing

The following preprocessing operations are considered for preparing the dataset.

1) Country and Language Subsetting: Relevant country and language subsets are selected for the experiments.

This allows controlled evaluation of the proposed framework across different cultural contexts.

2) Bilingual Alignment: Local-language and English question pairs are maintained together to ensure that both responses correspond to the same underlying cultural question.

3) Data Cleaning and Normalization: Duplicate entries, missing values, invalid samples, and formatting noise are handled during preprocessing.

Unicode and punctuation normalization is performed while preserving the original cultural meaning.

4) Question-Type Separation: Short-answer and multiple-choice questions are handled separately because their answer formats and evaluation requirements differ.

5) Knowledge Base Construction: Country-specific cultural information is organized into structured records containing:

- Country.
- Cultural category.
- Topic.
- Fact.
- Source.
- Relevant keywords and embeddings.

6) Embedding Generation and Indexing: Questions and cultural knowledge entries are converted into vector representations using an embedding model.

FAISS is used to construct a vector index for efficient similarity-based retrieval.

Metadata is retained alongside the vector representations to support country-specific filtering.

D. Data Augmentation

The following augmentation strategies are considered to improve the robustness of the evaluation framework.

- **Question Paraphrasing:**
  Generate semantically equivalent versions of cultural questions while preserving their original meaning.
- **Cross-Lingual Paraphrasing:**
  Create equivalent local-language and English question variants to evaluate robustness across different wording patterns.
- **Answer Perturbation and Conflict Cases:**
  Introduce controlled cases where local-language and English responses differ, allowing explicit evaluation of disagreement detection and knowledge-base-assisted adjudication.

These strategies are intended for controlled robustness evaluation and do not replace the original benchmark annotations.

E. Justification for Dataset Selection

BLEnD is suitable for this project because it provides a large collection of everyday cultural questions across multiple countries and languages.

Its bilingual question structure supports direct comparison between local-language and English responses.

Additionally, its combination of short-answer and multiple-choice questions allows the proposed system to be examined under different question-answering formats.

The benchmark also provides a useful basis for studying language-dependent cultural knowledge differences, which is the central motivation of this project.

F. Hardware Requirements

The expected hardware resources include:

- **GPU:** NVIDIA GPU with approximately 8–16 GB VRAM, such as RTX 3060, RTX 4060, RTX 4070, or equivalent.
- **CPU:** Intel i5 or AMD Ryzen 5 with at least four cores.
- **RAM:** Approximately 16 GB.
- **Storage:** 20–30 GB of available SSD storage.
- **Internet:** Required for external LLM API calls and online knowledge-source access.

G. Software and Frameworks

The following software tools and libraries are used or considered for the project:

- **Programming Language:** Python 3.10 or later.
- **Backend:** FastAPI and Uvicorn.
- **Data Processing:** Pandas, NumPy, JSON, SQLite3.
- **NLP and Embeddings:** Transformers and Sentence-Transformers.
- **Knowledge Retrieval:** FAISS, ChromaDB, and Rank-BM25.
- **Machine Learning:** Scikit-learn, SciPy, PyTorch, and Accelerate.
- **Visualization:** Matplotlib and Seaborn.
- **Development Environment:** Jupyter Notebook, VS Code, and Google Colab.
- **Version Control:** Git and GitHub.

H. LLM APIs and Embedding Models

External language models are used for response generation and adjudication.

The proposed and explored options include:

- GPT-based models.
- Gemini models through Google AI Studio.
- Llama models through Groq API.
- Other compatible instruction-tuned LLMs.

Embedding models considered include:

- multilingual-e5-large.
- all-MiniLM-L6-v2.
- BAAI/bge-large-en-v1.5.

The embedding model is responsible for representing cultural questions and knowledge entries in a common vector space for retrieval.

I. Resource Availability and Management

The project primarily relies on available institutional computing facilities, local development environments, and external LLM APIs.

When GPU resources are limited, embedding generation and retrieval operations can be performed using CPU-compatible implementations.

The use of pre-trained embedding models and external LLM APIs reduces the need for training large language models from scratch.

The FastAPI architecture also allows the embedding model and retrieval index to remain loaded in memory, reducing repeated initialization overhead.

V. Limitations Found

A. Limitations in Existing Approaches

The study of BLEnD and existing cultural question-answering approaches reveals several limitations relevant to this work.

1) **Language-Dependent Performance:**
   LLMs may generate different quality responses depending on whether a cultural question is presented in English or the corresponding local language.
2) **Single-Response Dependency:**
   Conventional systems often generate and return a single response without comparing alternative language-based outputs.
3) **Cross-Cultural Retrieval Errors:**
   Dense retrieval systems may retrieve semantically similar but culturally unrelated information from other countries.
4) **Inference Overhead:**
   Repeated initialization of embedding models and vector indexes can increase response latency.
5) **Limited Source Attribution:**
   Retrieved information may not always maintain sufficient source-level provenance for verification.
6) **Geographic Representation Bias:**
   Cultural information available for different regions may be unevenly represented in knowledge collections.

B. How the Proposed Work Addresses These Limitations

The project addresses the identified limitations through a combination of bilingual response generation, evidence retrieval, metadata filtering, and efficient system architecture.

C. Uniqueness of the Proposed Approach

The proposed approach differs from standard cultural question-answering implementations by combining several components into a unified inference-time framework.

A conventional system follows:

$$Q \rightarrow LLM \rightarrow A \tag{2}$$

The proposed framework follows:

TABLE II
RESEARCH LIMITATIONS AND PROPOSED IMPROVEMENTS

Research Limitation	Implemented Improvement	Intended Benefit
Cross-Cultural Semantic Bleed	Over-sampled FAISS retrieval with $k = 1000$ candidates followed by country-specific metadata filtering.	Reduces retrieval of culturally unrelated information.
Prototype Latency and Model Overhead	In-memory FastAPI microservice with cached embedding model and FAISS index.	Reduces repeated initialization and retrieval overhead.
Lack of Ground-Truth Attribution	Structured fact provenance using country, category, topic, and source URL.	Enables source-level traceability of retrieved knowledge.
Geographic Data Bias	Expansion of cultural knowledge collection across seven diverse countries.	Improves geographic diversity of the knowledge repository.

$$\begin{aligned} Q &\rightarrow (A_l, A_e) \\ &\rightarrow \text{Comparison} \\ &\rightarrow \text{Knowledge Retrieval} \\ &\rightarrow \text{LLM Judge} \\ &\rightarrow A_f \end{aligned} \tag{3}$$

where:

- Q represents the original question.
- A_l represents the local-language answer.
- A_e represents the English answer.
- A_f represents the final adjudicated answer.

The key distinction is the use of cross-lingual response comparison combined with external cultural evidence for final answer selection.

D. Expected Improvements

The project aims to improve the following aspects:

- **Performance:**
  Improve cultural question-answering accuracy, particularly in contexts where one language-based response may be insufficient.
- **Efficiency:**
  Avoid unnecessary adjudication when both language responses agree and retrieve only relevant knowledge.
- **Generalization:**
  Improve robustness across different languages, countries, question wordings, and cultural contexts.
- **Reliability:**
  Reduce unsupported or culturally inaccurate answers through evidence-based adjudication.
- **Traceability:**
  Maintain source information for retrieved cultural facts.

The effectiveness of these improvements must be established through systematic evaluation and comparison with appropriate baselines.

VI. EXPECTED OUTCOMES

A. Project Goals

The primary goals of the project are:

1) Develop a dual-language ensemble pipeline that generates responses in both local language and English.
2) Implement a disagreement detection mechanism to identify cases where the two responses differ.
3) Build a structured Cultural Knowledge Base containing country-specific cultural information.
4) Implement a retrieval mechanism to fetch relevant cultural evidence for a given question.
5) Develop an LLM-as-a-Judge module that uses both candidate answers and retrieved evidence to select the final response.
6) Evaluate the framework on BLEnD and the selected additional cultural knowledge collections.
7) Analyze whether knowledge-base-assisted adjudication improves answer quality and reduces language-dependent performance differences.

B. Performance Evaluation Metrics

The proposed framework will be evaluated using the following metrics.

1) Primary Metrics:

- **Accuracy:**
  Percentage of correctly answered cultural questions.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100 \tag{4}$$

- **Accuracy Improvement Over Baselines:**
  Performance improvement relative to local-language-only and English-only systems.

$$\Delta A = A_f - A_b \tag{5}$$

  where A_f is the final system accuracy and A_b is the baseline accuracy.
- **Cultural Knowledge Gap Reduction:**
  Difference between the best-performing and worst-performing language or country settings.

$$G = A_{\max} - A_{\min} \tag{6}$$

  A reduction in this gap after applying the proposed framework would indicate more balanced performance.

2) Secondary Metrics:

- **Adjudication Success Rate:**
  Percentage of disagreement cases where the judge produces a more accurate final answer than either individual response.
- **Knowledge Retrieval Relevance:**
  Fraction of retrieved knowledge-base entries considered relevant to the corresponding cultural question.
- **Response Latency:**

TABLE III
TENTATIVE EVALUATION RECORD SCHEMA

Record ID	Country/Region	Local Language	Resource Level	Cultural Category	Question Type	Question	Ground Truth	Local Answer	English Answer
Generated ID	Dataset Value	Dataset Value	Dataset Value	Dataset Value	SAQ/MCQ	BLEnD Question	Reference Answer	Model Output	Model Output

TABLE IV
TENTATIVE EVALUATION RECORD SCHEMA (CONTINUED)

Similarity	Agreement	Disagreement	Retrieved Evidence	Evidence Relevance	Judge Decision	Final Answer	Final Correct	KB Grounded	Latency
Calculated	Agree/Partial/Disagree	Yes/No	Retrieved KB Facts	Calculated	Judge Output	Final Output	Evaluated	Yes/No	Measured

Average time required to generate the final response.

- **Knowledge Base Grounding:**
  Percentage of final responses supported by retrieved cultural evidence.

C. Evaluation Record Schema

The following tables define the proposed structure for storing and analyzing experimental records.

D. Expected Deliverables

The expected project deliverables are:

1) **D1. Cultural Knowledge Base**
   A structured repository of country-specific cultural facts, customs, traditions, foods, festivals, and everyday practices.
2) **D2. Dual-Language Ensemble System**
   An end-to-end pipeline that generates local-language and English responses, compares them, and identifies disagreements.
3) **D3. Knowledge-Base-Assisted LLM-as-a-Judge Module**
   A decision-making component that selects or synthesizes the final answer using the generated responses and retrieved cultural evidence.
4) **D4. Evaluation Results and Analysis**
   Accuracy comparisons, language-wise and country-wise performance analysis, cultural gap analysis, and visualizations.
5) **D5. FastAPI-Based Demonstration System**
   A working API or application demonstrating the complete workflow:
   $$\text{Question Input} \rightarrow \text{Dual Responses} \rightarrow \text{Knowledge Retrieval} \rightarrow \text{LLM Judge} \rightarrow \text{Final Answer}$$
6) **D6. Final Research Report**
   A complete technical report documenting the methodology, implementation, experiments, results, limitations, and future research directions.

E. Conclusion and Future Scope

This project investigates a framework for improving cultural question answering by combining bilingual response generation with evidence-based answer adjudication.

By utilizing both local-language and English responses, the system aims to reduce dependence on a single language representation.

The integration of a structured Cultural Knowledge Base provides an additional source of information for verifying generated responses.

Furthermore, metadata-scoped retrieval, in-memory inference infrastructure, structured fact provenance, and geographic knowledge expansion address important practical limitations in cultural retrieval systems.

Future work may explore more advanced cross-lingual embedding models, adaptive retrieval strategies, confidence-aware adjudication, larger cultural knowledge collections, and evaluation across additional low-resource languages and regions.

REFERENCES

[1] J. Myung, N. Lee, Y. Zhou, J. Jin, R. A. Putri, D. Antypas, H. Borkakoty, E. Kim, C. Perez-Almendros, A. A. Ayele, V. Gutiérrez-Basulto, Y. Ibáñez-García, H. Lee, S. H. Muhammad, K. Park, A. S. Rzayev, N. White, S. M. Yimam, M. T. Pilehvar, N. Ousidhoum, J. Camacho-Collados, and A. Oh, "BLEnD: A Benchmark for LLMs on Everyday Knowledge in Diverse Cultures and Languages," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.
[Online]. Available: https://arxiv.org/abs/2406.09948

[2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W. Yih, T. Rocktaschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.

[3] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems*, 2023.

[4] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 3982–3992, 2019.

[5] J. Johnson, M. Douze, and H. Jégou, "Billion-scale Similarity Search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

APPENDIX

The following tools and platforms are used or considered during the development and documentation of the project.

- **Programming and Development:** Python, VS Code, Jupyter Notebook, Google Colab.
- **LLM Platforms:** Google AI Studio (Gemini), Groq API (Llama), and other compatible LLM APIs.

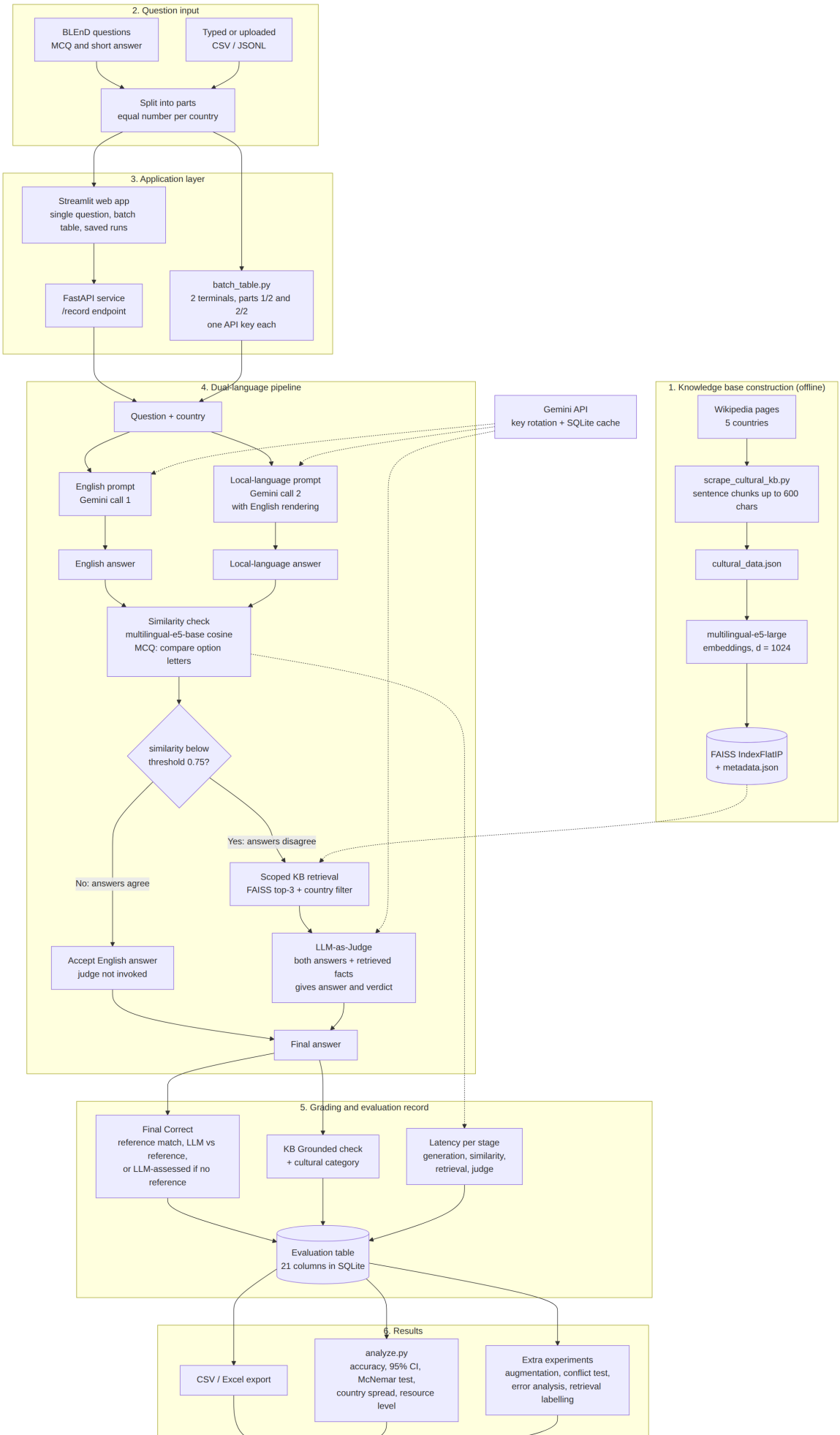
- **Backend Development:** FastAPI and Uvicorn.
- **Knowledge Retrieval:** FAISS and Sentence-Transformers.
- **Data Management:** SQLite, JSON, Pandas, and NumPy.
- **Version Control:** Git and GitHub.
- **Dataset Resources:** BLEnD GitHub repository, Hugging Face, and public cultural information sources.

A. Use of Generative AI Tools

Generative AI tools may be used as development and documentation aids for brainstorming, understanding technical concepts, debugging, and improving the clarity of written content.

The project methodology, implementation, experimental decisions, and final research conclusions must be reviewed and verified by the authors.

Any generated content used in the final submission should be checked against the corresponding technical implementation and research sources.


2. Question input
BLEnD questions
MCQ and short answer
Typed or uploaded
CSV / JSONL
Split into parts
equal number per country
3. Application layer
Streamlit web app
single question, batch
table, saved runs
FastAPI service
/record endpoint
batch_table.py
2 terminals, parts 1/2 and
2/2
one API key each
4. Dual-language pipeline
Question + country
Gemini API
key rotation + SQLite cache
English prompt
Gemini call 1
Local-language prompt
Gemini call 2
with English rendering
English answer
Local-language answer
Similarity check
multilingual-e5-base cosine
MCQ: compare option
letters
similarity below
threshold 0.75?
Yes: answers disagree
No: answers agree
Scoped KB retrieval
FAISS top-3 + country filter
Accept English answer
judge not invoked
LLM-as-Judge
both answers + retrieved
facts
gives answer and verdict
Final answer
1. Knowledge base construction (offline)
Wikipedia pages
5 countries
scrape_cultural_kb.py
sentence chunks up to 600
chars
cultural_data.json
multilingual-e5-large
embeddings, d = 1024
FAISS IndexFlatIP
+ metadata.json
5. Grading and evaluation record
Final Correct
reference match, LLM vs
reference,
or LLM-assessed if no
reference
KB Grounded check
+ cultural category
Latency per stage
generation, similarity,
retrieval, judge
Evaluation table
21 columns in SQLite
6. Results
CSV / Excel export
analyze.py
accuracy, 95% CI,
McNemar test,
country spread, resource
level
Extra experiments
augmentation, conflict test,
error analysis, retrieval
labelling